JNTUH USED 02-12-17 PM Set-1

Code No: 117EF **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, November/December - 2017
MACHINE LEARNING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Define learning. [2]
b) What is the influence of information theory on machine learning? [3]
c) What is an active query? [2]
d) List the characteristics of problems for which decision tree learning is best suited. [3]
e) State Baye's theorem. [2]
f) Differentiate between sample error and true error. [3]
g) What are the disadvantages of instance based methods? [2]
h) What is the inductive bias of k-nearest neighbor? [3]
i) What is analytical learning? [2]
j) List the factors motivated the popularity of genetic algorithms. [3]

PART-B

(50 Marks)

2. What is meant by machine learning? What is its need to today's society? Explain successful applications of machine learning. [10]

OR

3.a) Illustrate general-to-specific ordering of hypotheses in concept learning.
b) Explain the key property of FIND-S algorithm for concept learning with necessary example. [5+5]

4. Present the basic ID3 algorithm for learning decision trees and illustrate its operation in detail. [10]

OR

5.a) Discuss the representational power of a perceptron.
b) Explain the gradient descent algorithm for training a linear unit. Implement stochastic approximation to this. [5+5]

6.a) Describe a general approach for deriving confidence intervals.
b) Explain the features of Bayesian learning methods. [5+5]

OR

7.a) With an illustrative example explain brute force MAP learning algorithm.
b) What are the applications of probably approximately correct model? Discuss in detail. [5+5]

8.a) Demonstrate k-nearest neighbor algorithm for classification.
b) Discuss the significance of locally weighted regression. [5+5]

OR

9. Explain how CADET system employs case based reasoning to assist in the conceptual design of simple mechanical devices. [10]

10. Consider the two strings as initial population for genetic algorithm and generate all possible off springs using various operators.
String 1: 11101001000
String 2: 00001010101 [10]

OR

11.a) What are the main properties of PROLOG-EBG algorithm? Is it deductive or inductive? Justify your answer.
b) Write KBANN algorithm to explain usage of prior knowledge to reduce complexity. [5+5]

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